

Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

SOFT COMPUTING PARADIGMS [CSE-4054]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A neural network's single neuron receives inputs of (0.25, 0.55, 0.35) with corresponding weights of (3.0, 2.5, 1.0) and a bias of 1.2. The neuron's activation function is the ReLU function, defined as:
 $f(x) = \max(0, x)$
Given an actual output of 1.05, solve the following:
- A) i) Calculate the neuron's output after applying the activation function. (4)
ii) Draw the architecture of the neuron.
iii) Calculate the loss (error) after the feedforward step using the MAE loss function, $MAE = |Actual - Predicted|$.
iv) Suggest the next step to minimize the error.
- B) A financial institution is developing a system to assess the creditworthiness of loan applicants. The system can be designed using two different approaches: Hard computing approach and soft computing approach. Explain how both the hard computing and soft computing approaches could be applied to the creditworthiness assessment system. Discuss the advantages and limitations of each approach in this scenario. (3)
- C) (3)

A retailer wants to group its customers into 2 clusters based on their annual income and the number of purchases they made in the past year using the K-means clustering algorithm. The dataset includes the following points:

Customer	Annual Income	Number of Purchases
A	20	30
B	35	20
C	50	40
D	70	25
E	55	35
F	80	50

Perform one iteration of the K-means clustering algorithm with $k = 2$, starting with the following initial centroids:

- a. Centroid 1: (Annual Income = 20, Purchases = 30)
- b. Centroid 2: (Annual Income = 80, Purchases = 50)

After assigning the points to clusters, update the centroids based on the new clusters (The distance is calculated using Euclidean formula).

- 2) Analyze how the neighborhood function influences the learning process in a Self-Organizing Map (SOM). Discuss how the size of the neighborhood changes over time and its impact on the organization of neurons. (5)
 - A)
 - B) Given a 4x4 Self-Organizing Map and an input vector [0.3, 0.9], with the BMU's current weight vector [0.4, 0.8] and a learning rate of 0.05, perform a weight update for the BMU. (5)
- 3) Analyze the basic components of a Fuzzy Inference System (FIS). Discuss the role of fuzzification, rule base, inference engine, and defuzzification in the FIS process. (5)
 - A)
 - B) Consider a Fuzzy Inference System with two inputs, each having three fuzzy sets (Low, Medium, High). If you have the following rules:
 Rule 1: IF Input1 is Low AND Input2 is High, THEN Output is Medium.
 Rule 2: IF Input1 is Medium AND Input2 is Low, THEN Output is High.
 Given Input1 = 0.3 and Input2 = 0.8, calculate the firing strength of Rule 1 using the minimum method of aggregation (min). (5)
 Assume the fuzzy membership values for Low at Input1 = 0.3 and High at Input2 = 0.8 are 0.6 and 0.7, respectively.
- 4) Given a scenario suppose that the size of the chromosome population $N=6$ and the fitness function is $15x-x^2$ and the binary code for the string is 1100, 0100, 0001, 1110, 0111, 1001 here X in the fitness function is the decimal value of the given string. perform the following: (5)
 - A)

- i) Find the fitness values of the strings and draw the Roulette wheel.
- ii) After performing (i) do the first iteration in which crossover is done between 6th string and 2nd string at third position, onwards. Calculate and show the new fitness values.

- B) How does ACO differ from other optimization techniques like Genetic Algorithms or Particle Swarm Optimization? (5)
- 5) Compare and contrast different encoding techniques used in Evolutionary Neural Networks. (5)
 - A)
 - B) Analyze the PSO algorithm for following scenario:
The fitness function used by the problem is $f(x,y) = x^2 + y^2$, where x and y are the dimensions of the problem, the velocities of all particles are initialized to zero and inertia $w=0.3$ and the value of the cognitive and social constants are, $c_1=2$ and $c_2=2$. (5)
Assume that the particles are the following initial values. $p_1(1,1)$, $p_2(-1,1)$, $p_3(0.5,-0.5)$, $p_4(1,-1)$, $p_5(0.25,0.25)$. Show the fitness and position computations for two iterations

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